

EstateIQ

REAL ESTATE INVESTMENT ADVISOR

Real Estate Investment Insights Report

Predicting property profitability and future value across 250,000 Indian listings, with SQL analytics, machine learning and an interactive Streamlit advisor.

250,000	33	88%	39%
listings analysed	SQL queries	classifier accuracy	flagged Good Investment

Prepared 26 October 2026 · Python · SQLite · scikit-learn · Streamlit

1. Executive summary

This report answers two investor questions: **Is this property a good investment?** and **What will it be worth in five years?** It combines 33 SQL queries on the full dataset with two machine-learning models.

Five findings that matter

1. Entry price versus the city median is the biggest lever. Properties priced below their city's median ₹/sq ft qualify as Good Investments far more often (81% for deep discounts, 55% below median) than premium-priced ones (0%).

2. Descriptive price data is remarkably flat. State, city, property type, furnishing, facing, schools and hospitals all move average price by under 5% (state 2.5%, city 4.8%, the rest under 1%). Location alone will not tell an investor where to buy.

3. Size is priced per rupee, not per square foot. Total price is unrelated to size (r = -0.00) so bigger homes are far cheaper per sq ft (r = -0.61).

4. Under-construction, well-connected, amenity-rich homes score highest. Under-construction listings qualify 49% of the time versus 30% for ready-to-move; five amenities lift the rate from 30% to 48%.

5. The models learn the investment rules well. The classifier reaches 87.6% accuracy (ROC-AUC 0.953); the 5-year price regressor has RMSE 16.1 lakhs (R^2 0.994), beating a constant-growth baseline of 34.9.

> **Read this first.** The raw file contains no price history and no investment label. Annual appreciation, 5-year price and Good Investment are **modelled** from transparent domain rules (Section 2). Findings tagged **(modelled)** reflect those rules; everything else is observed in the data.

Recommendations

- Screen on **price vs city median** first, then infrastructure score, transit access and amenities.
- Prefer **larger units** for value per sq ft and **under-construction** stock for modelled upside.
- Use **medians**, not averages, when quoting prices: price/sq ft has a long right tail.
- Fix data capture: about 46% of rows list a floor above the building's height.
- Replace the rule engine with real transaction history to turn modelled upside into a true forecast.

2. Data and methodology

The dataset covers 250,000 listings across 20 states, 42 cities and 500 localities. The pipeline (`scripts/run_pipeline.py`) cleans the data, engineers features, loads SQLite and trains the models.

Data-quality results (SQL Q01-Q03)

Check	Result
Rows / unique IDs	250,000 / 250,000
Missing values (price, size, city)	0
Duplicate rows	0
Floor_No above Total_Floors	116,304 rows (46.5%) - flagged, floor ratio clipped
Price/sq ft outliers (IQR rule)	19,724 rows (7.89%) - flagged, kept
Size outliers	0

Engineered features

Price per sq ft in rupees, Amenity_Count, Transport_Score, School_Density_Score, Infra_Score (0-100), Floor_Ratio, Value_Ratio (price/sq ft ÷ city median), parking/security flags, age and price bands. Numeric features are standardised and categoricals one-hot encoded inside the model pipeline.

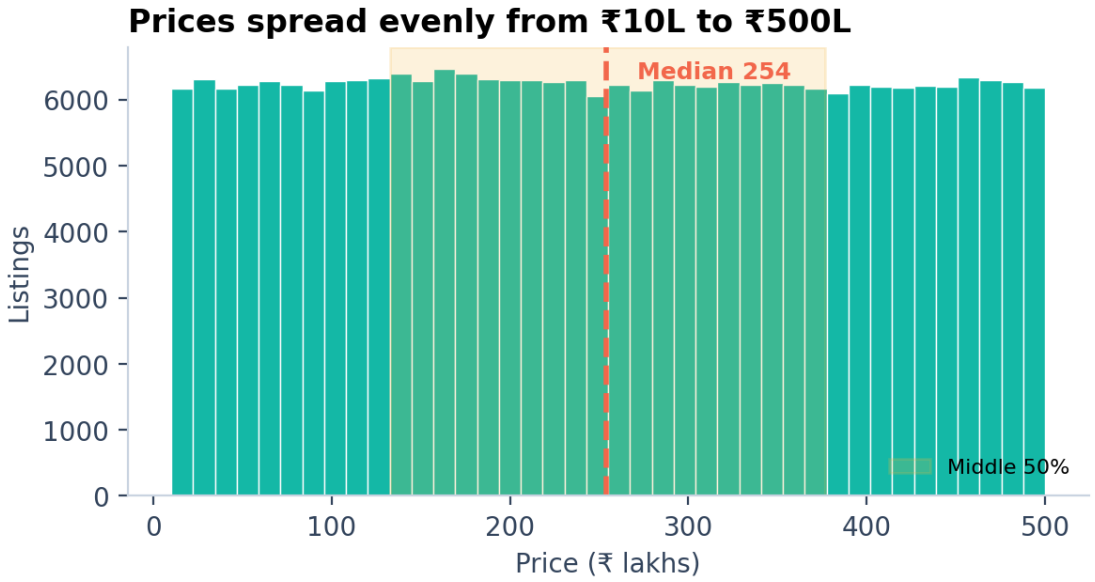
How the investment targets are defined

Component	Effect on annual appreciation
Base rate	4.5%
Infrastructure (schools, hospitals, transit)	up to +3.0%
Amenities / security / parking	up to +1.2% / +0.8% / +0.6%
Under construction / newer building	+1.0% / up to +1.0%
Price vs city median (cheaper is better)	-2.0% to +2.0%
Simulated market noise	N(0, 0.8%)

5-year price = price × $(1 + \text{appreciation})^5$. **Good Investment** = appreciation ≥ 9% and price/sq ft ≤ 1.5× the city median. 39.2% of listings qualify.

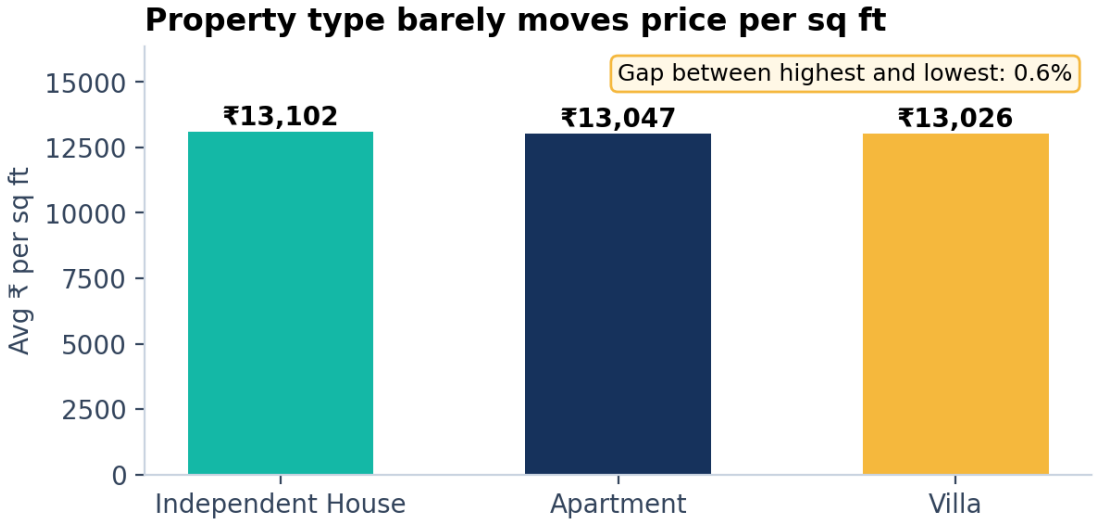
3. Price and size analysis

SQL queries Q04-Q08 · EDA questions 1-5



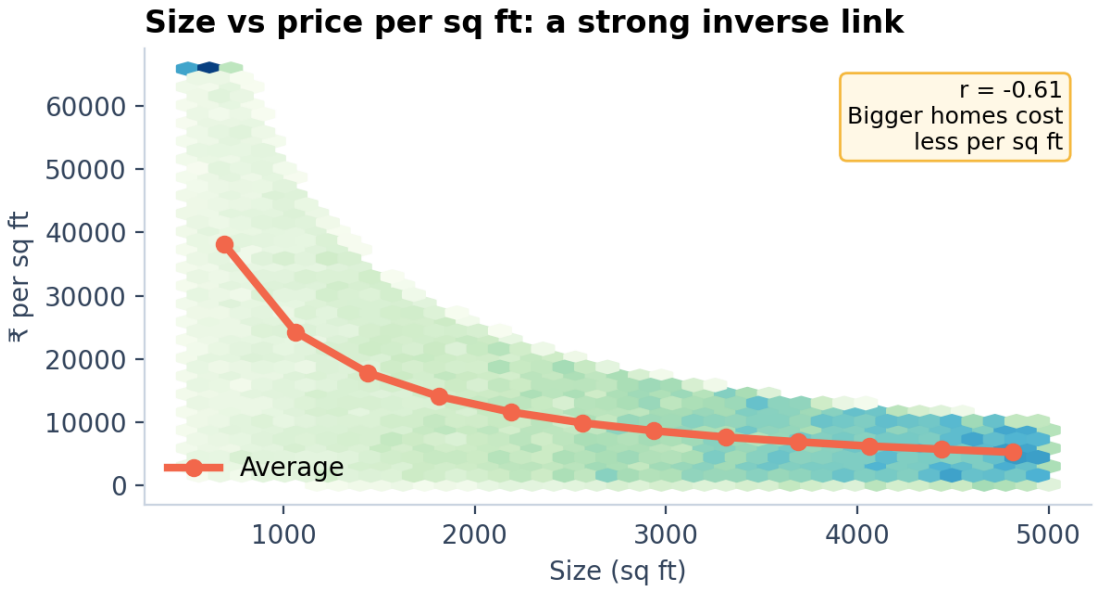
Q1 · Distribution of property prices

Prices run evenly from ₹10L to ₹500L (median ₹254L, mean ₹255L). There is no single typical tier, so segment by budget before comparing properties.



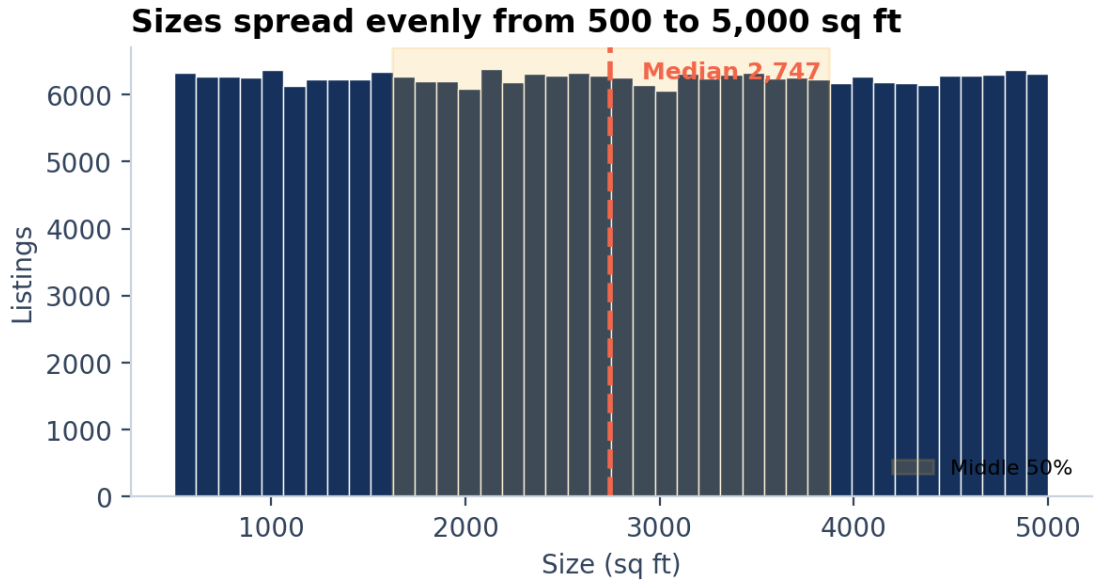
Q3 · Price per sq ft by property type (SQL Q06)

Villas, apartments and independent houses price within 0.6% of each other per sq ft. Do not pay a premium for the label alone.



Q4 · Size versus price per sq ft

Correlation r = -0.61. Total price is unrelated to size (r = -0.00), so larger homes are materially cheaper per sq ft: the best value per square foot sits in big units.



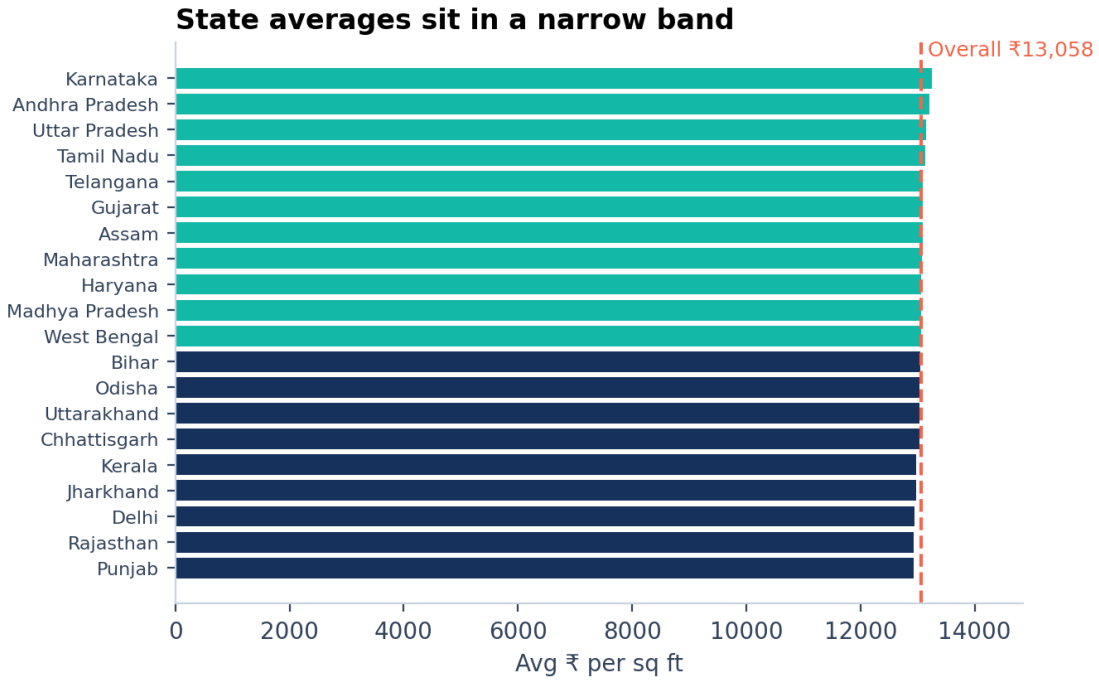
Q2 · Distribution of property sizes

Every 500 sq ft band holds a similar share of listings, so supply is not skewed toward compact homes.

Q5 · Outliers. 7.89% of listings (19,724) are price-per-sq-ft outliers, all on the high side (small footprint, high total price). Size has no outliers. Use medians for pricing decisions.

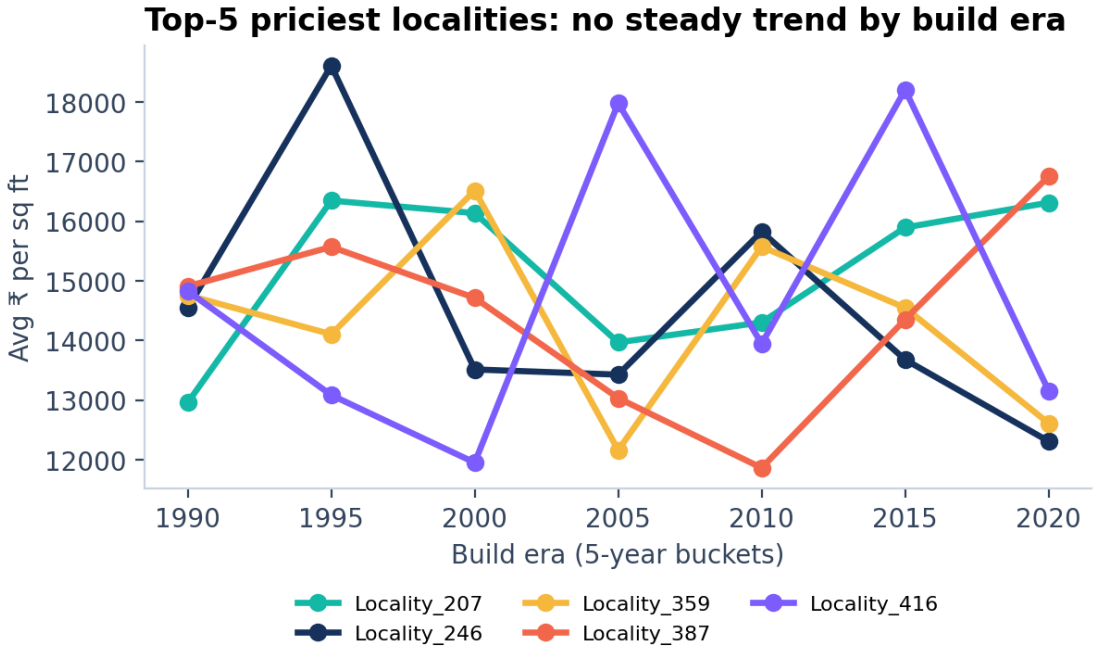
4. Location analysis

SQL queries Q09-Q13 · EDA questions 6-10



Q6 · Average price per sq ft by state (SQL Q09)

The highest state (Karnataka, ₹13,252) and the lowest (Punjab, ₹12,931) are only 2.5% apart. Location alone does not explain price in this data.



Q10 · Price trend of the five most expensive localities (SQL Q13)

Premium localities show no steady rise by build era; newer is not automatically pricier. Evaluate each locality individually.

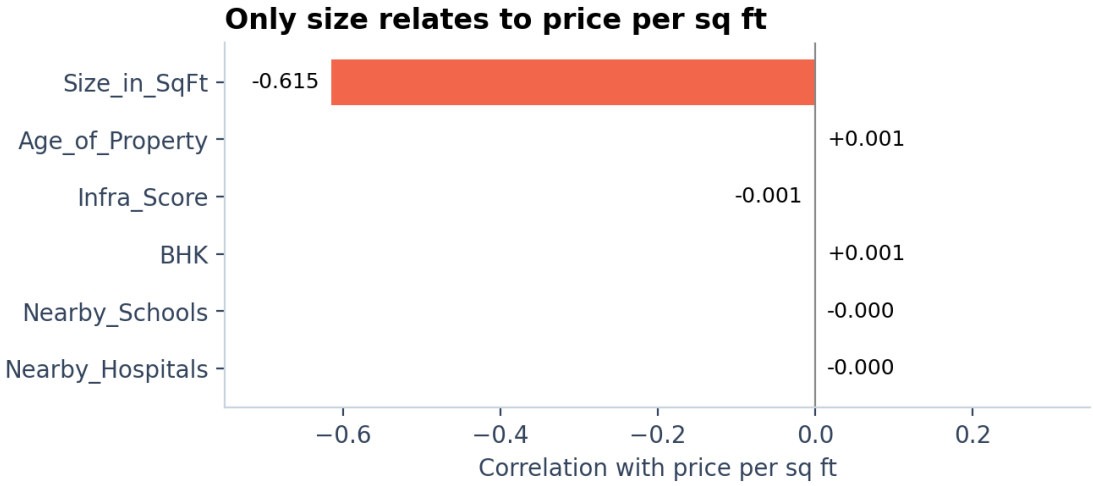
Q7 · Top five cities by average price (SQL Q10)

City	State	Listings	Avg price (L)	Avg Rs/sq ft
Bangalore	Karnataka	4,144	258.5	13,140
Surat	Gujarat	6,167	258.1	13,377
Kochi	Kerala	6,305	257.4	13,203
Gaya	Bihar	6,170	257.1	13,146
Dehradun	Uttarakhand	6,234	256.9	13,006

City averages differ by only a few lakhs (about 3% from top to bottom of the full list); entry price against the local median matters more than the city.

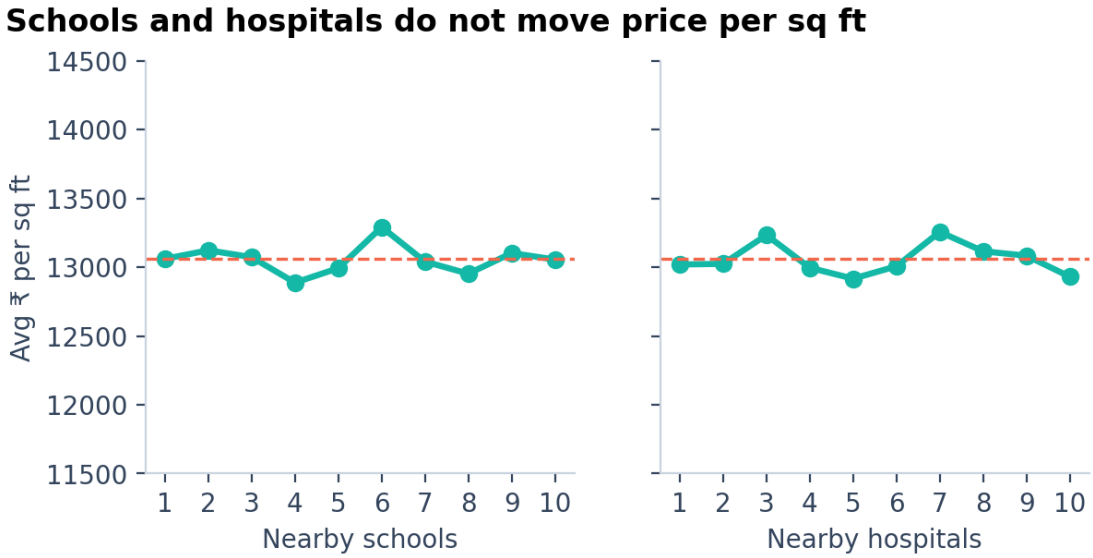
5. Feature relationships

SQL queries Q14-Q18 · EDA questions 11-15



Q11 · Correlation with price per sq ft (SQL Q14)

Size is the only feature meaningfully related to price per sq ft. Age, BHK, schools, hospitals and infrastructure score all sit at ~0.



Q12-13 · Nearby schools and hospitals versus price per sq ft (SQL Q15, Q16)

Both lines are flat around the overall average (dashed): schools and hospitals are livability factors, not price drivers, in this dataset.

Q14-15 · Furnishing and facing (SQL Q17, Q18)

Furnishing	Avg price (L)	Avg Rs/sq ft
Unfurnished	255.0	13,048
Furnished	254.4	13,034
Semi-furnished	254.3	13,093

Facing	Avg Rs/sq ft	% vs overall
West	13,140	+0.63%
South	13,044	-0.11%
North	13,025	-0.26%
East	13,024	-0.26%

Furnishing changes asking price by under 0.3% and facing direction by under 1%. A fully furnished home is effectively a free upgrade, and there is no visible Vastu premium.

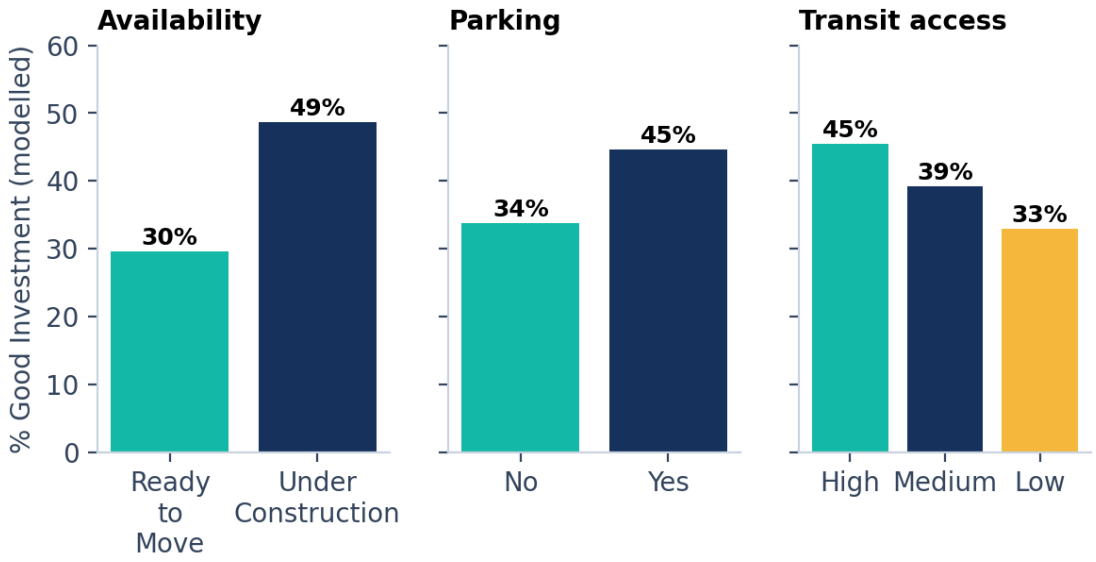
6. Investment, amenities and ownership

SQL queries Q19-Q24 · EDA questions 16-20

Q16 · Properties by owner type (SQL Q19)

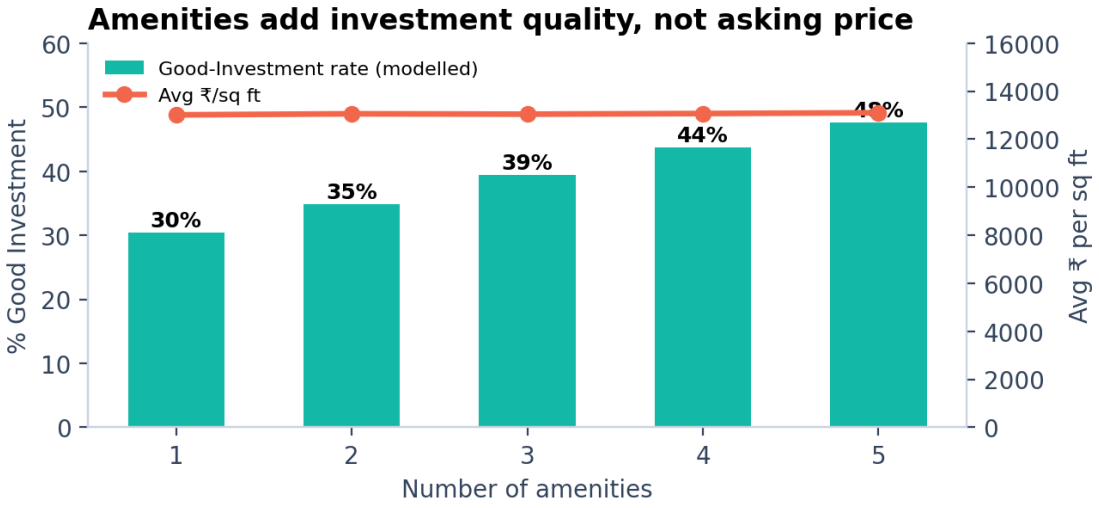
Owner type	Listings	Share	Good-Investment rate
Broker	83,479	33.4%	39.1%
Owner	83,268	33.3%	39.2%
Builder	83,253	33.3%	39.3%

Q17 · Availability status: 50% Ready-to-Move, 50% Under-Construction (SQL Q20).



Q17-20 · Good-Investment rate by availability, parking and transit access (modelled; SQL Q20, Q21, Q24)

Under-construction homes qualify 49% of the time (vs 30%); parking adds about 11 points; high transit access reaches 45% (vs 33% for low). None of these change the asking price, so they are cheap sources of upside. (modelled)

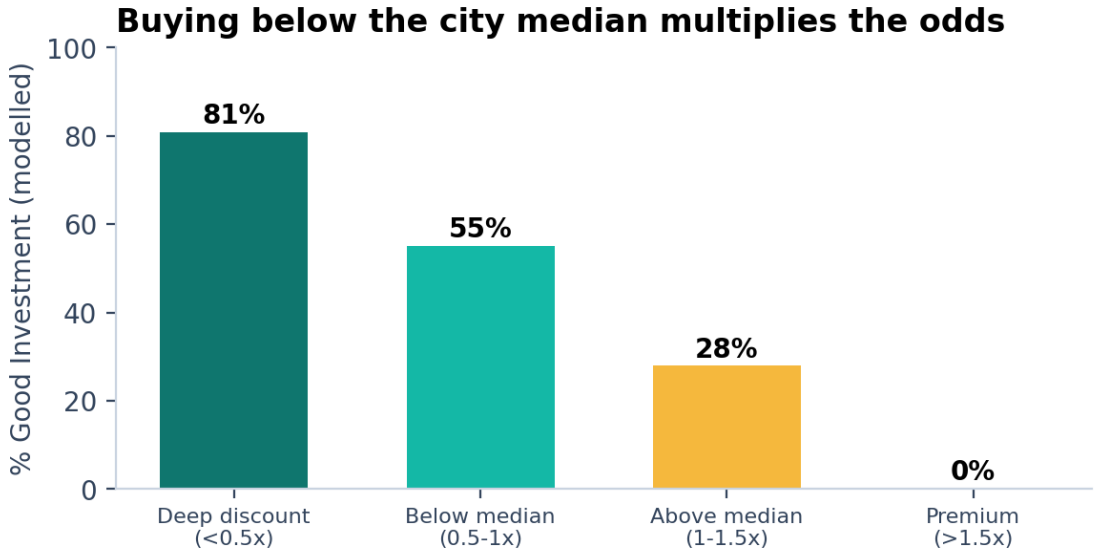


Q19 · Amenity count versus price and investment quality (SQL Q22)

Price per sq ft is flat across amenity counts, but the modelled Good-Investment rate rises from 30% with one amenity to 48% with five. (modelled)

7. Investment deep-dive

SQL queries Q25-Q33 · window functions, CTEs, ranking



Good-Investment rate by valuation band (SQL Q26)

Deep discounts qualify 81% of the time; premium-priced homes (above 1.5× the city median) never do, by definition of the rule. Negotiating the entry price is the single strongest lever. (modelled)

Top best-buy shortlist (SQL Q27 - first six rows; growth is capped at 14%)

ID	City	Type	BHK	Price (L)	5-yr (L)	Growth	vs median
99484	Jodhpur	Apartment	1	75	145	14.0%	0.28x
62939	Ahmedabad	Independent	2	328	632	14.0%	0.91x
230074	Patna	Apartment	3	132	254	14.0%	0.34x
67793	Surat	Apartment	1	56	107	14.0%	0.16x
67976	Pune	Independent	1	199	383	14.0%	0.53x
113553	Jaipur	Independent	5	70	135	14.0%	0.18x

SQL spotlight: valuation bands

```
SELECT CASE WHEN Value_Ratio < 0.5  THEN '1. Deep discount (<0.5x)'
           WHEN Value_Ratio < 1.0  THEN '2. Below median (0.5-1x)'
           WHEN Value_Ratio < 1.5  THEN '3. Above median (1-1.5x)'
           ELSE                         '4. Premium (>1.5x)' END AS valuation_band,
      COUNT(*) AS properties,
      ROUND(100.0 * AVG(Good_Investment), 1)       AS good_investment_pct,
      ROUND(100.0 * AVG(Annual_Appreciation), 2)   AS avg_appreciation_pct,
      ROUND(AVG(Projected_Gain_Lakhs), 1)          AS avg_gain_lakhs
FROM properties
GROUP BY valuation_band
ORDER BY valuation_band;
```

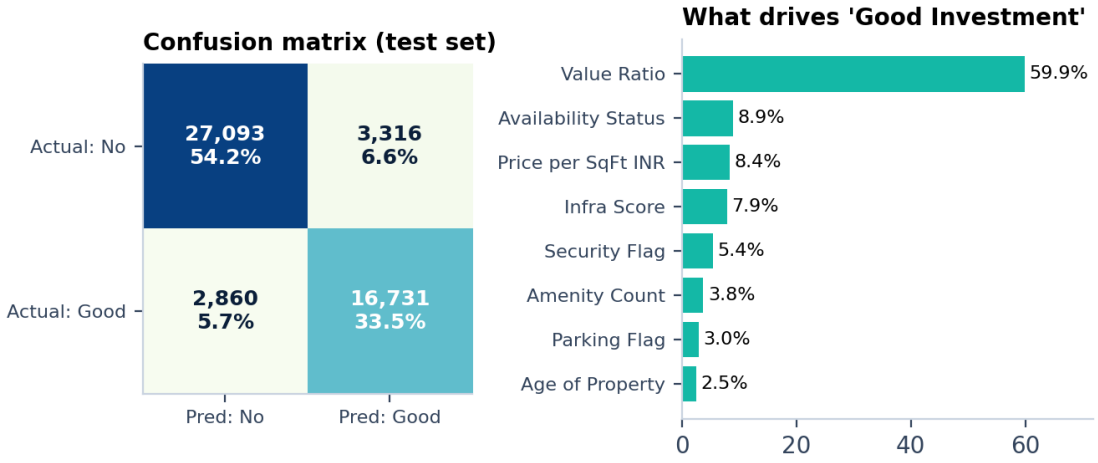
> Percentage returns fall as price rises (cheaper bands sit below their city median) while rupee gains rise: the cheapest band projects ~10.1% a year, the 400L+ band ~7.4%.

8. Machine-learning results

Three models were compared for each task on a stratified 80/20 split; experiments are tracked with MLflow when installed. **Gradient Boosting** won classification and **Gradient Boosting** won regression.

Classifier	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.873	0.835	0.842	0.839	0.951
Random Forest	0.863	0.817	0.837	0.827	0.944
Gradient Boosting	0.876	0.835	0.854	0.844	0.953

Regressor	RMSE (L)	MAE (L)	R²
Ridge Regression	20.88	15.97	0.9891
Random Forest	17.85	12.66	0.9921
Gradient Boosting	16.09	11.38	0.9936
Naive baseline (constant growth)	34.87	-	-



Classifier diagnostics and drivers

The classifier correctly labels 87.6% of unseen listings. Valuation versus the city median accounts for about 60% of its predictive power, followed by availability status, price per sq ft and infrastructure.

Interpretation. Targets are rule-derived plus simulated noise, so the scores show how well the models learn the investment logic from property features, not accuracy on real sale prices. The regressor's RMSE of 16.1L is a 54% improvement over a constant-growth baseline.

9. Recommendations and limitations

For investors

1. **Negotiate to below the city median.** The Streamlit advisor shows the price at which a property turns into a Good Investment.
2. **Favour infrastructure and transit** over a prestige location: they cost no premium in this data.
3. **Prefer larger units** for lower cost per sq ft.
4. **Look at under-construction stock** for the highest modelled upside, and weigh the delivery risk.

For real-estate platforms

Show 'vs city median' beside every listing, surface the Good-Investment flag and confidence score, and fix floor-number validation at data entry.

Limitations

• No price history: appreciation and 5-year price are scenario-based.
• Descriptive data shows almost no link between price and location or amenities, typical of synthetic data.
• No crime-rate field, so the crime analysis in the brief could not be done.
• Not financial advice.

Deliverables

Item	File
SQL analysis (33 queries)	sql/real_estate_analysis.sql
Streamlit app + SQL playground	app.py, views/, src/
Cleaned dataset	data/housing_processed.csv
Training pipeline + MLflow	scripts/run_pipeline.py, src/modeling.py
Logo	assets/logo.png / .svg
Presentation	EstateIQ_Presentation.pptx